Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eighth Semester of B. Tech. (CL) Examination

April-May 2018

CL 406.01 Design of Structures-II

Date: 04.05.2018, Friday **Time: 01.30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed
5. IS 456:2000, IS 800:2007, IS 875:1987 (Part I-V), SP 16, SP 6, SP 34 and steel table are permitted, provided that they do not contain anything other than the printed matter inside.
6. Assume Fe 415 and M 20 unless otherwise mentioned.

SECTION – I

Q-1 Design the staircase shown in **Figure 1**. The risers are 15 cm and goings are 25 cm, and story height is 3.5 m. Goings are provided with 3cm-thick marble finish on cement mortar that weighs 120 kg/m^2, while 1.5 cm thick plaster is applied to both the risers and bottom surfaces of the slab. The stair is to be designed for a live load of 250 kg/m^2. **[07]**

Q-2 A typical floor plan is shown in **Figure 2** has the following data: **[14]**

(i) Type of structure: Multi-storey frame

(ii) Number of storeys: 4 (G + 3)

(iii) Floor to floor height: 5 m

(iv) Height of plinth: 0.6 m above G.L.

(v) Live Load: 3 kN/m^2 at typical floor & 1.5 kN/m^2 on terrace

(vi) Floor finish: 1 kN/m^2, Water proofing: 2 kN/m^2, Terrace Finish: 1 kN/m^2

(vii) Walls : 230 mm thick brick masonry walls only at periphery

(viii) Size of main beams: 230 mm × 300 mm

(ix) Size of secondary beams: 230 mm × 230 mm

Take M25 grade concrete and Fe415 steel.

Based on the above data, attempt **any TWO** of the following.

(a) Design and Detail Two way panel slab

(b) Total load on all main beams and floor beams

(c) Design and detail One way panel slab

Q-3 (a) Determine the plan dimensions of a combined footing for two axially loaded columns C_1 and C_2 having size 400 mm x 400 mm and 300 mm x 500 mm respectively. The column C_1 is flushed with the property line and center to center distance between both the columns is 3 m. Column C_1 and C_2 carry factored load of 800 kN and 1200 kN respectively. Safe bearing capacity of sub soil is 150 kN/m^2. **[04]**

Q-3 (b) Draw qualitative deflection shape of cantilever retaining wall. Design the cross section of a cantilever retaining wall (T type) to retain earth for a height of 4 m above ground level. The back fill is horizontal. The density of soil is 18 kN/m^3. **[10]**

Safe bearing capacity of soil is 200 kN/m^2. Take the co-efficient to friction between concrete and soil as 0.6. The angle of repose is 30°. Use M20 concrete and Fe415 steel. Also do necessary checks for stability of retaining wall.

OR

(b) Design the cross section and stem of a counterfort retaining wall to retain earth for a height of 6 m. The density of soil is 16 kN/m^3. Safe bearing capacity of soil is 160 kN/m^2. Take the co-efficient to friction between concrete and soil as 0.67. The angle of repose is 30°. Surcharge pressure is 16 kN/m^2. Use M20 concrete and Fe415 steel. **[10]**

SECTION – II

Q-4 Calculate nodal Dead load, Live load and wind load for roof truss of an industrial building as shown in **Figure 3** with 15 m span and 40 m long. The roofing is galvanized iron sheeting. Spacing between two consecutive trusses is 5 m. The basic wind speed is 50 m/s and terrain is open industrial area and building is class A building. The building clear height at the eaves is 7 m. **[07]**

Q-5 (a) Design a suitable slab base for a column section ISHB 200 @ 365.9 N/m supporting an axial load of 400 kN. The base plate is to rest on a concrete pedestal of M 20 grade. Use Fe 410 grade of steel. **[06]**

(b) Design a welded plate girder 24 m in span and laterally restrained throughout. It has to support a uniform load of 100 kN/m throughout the span exclusive of self-weight. Design the girder using intermediate transverse stiffeners. The steel for the flange and web plates is of grade Fe 410. Design the cross section and end loading stiffener. **[08]**

OR

(b) Design a welded plate girder of span 20 m with transverse stiffeners using the tension field action for the following factored forces. **[08]**

M_z = 5000 kNm

Shear Force = 900 kN

The girder is laterally restrained. Connections Need not to be designed.

Q-6 (a) Derive the equation for optimum depth and web thickness of plate girder. **[06]**

OR

(a) Design the purlin for the following data: **[06]**

Span of purlin = 4 m

Spacing of purlin = 1.275 m

Angle θ = 11.3°

Live load = 0.724 kN/m^2, Dead load = 0.21 kN/m^2 , Wind pressure = 25.86 kN/m^2

Q-6 (b) Design the gantry girder for the following data: **[08]**

Type of Crane: MOT

Maximum hook approach: 0.8 m

Crane capacity: 300 kN

Span of Gantry: 10 m

Crab weight: 60 kN

Span of crane girder: 30 m

Wheel base width: 4 m

Gantry rail self-weight: 400 N/m
Diameter of wheel: 200 mm
Life span in years: 50
Grade of steel: Fe 410

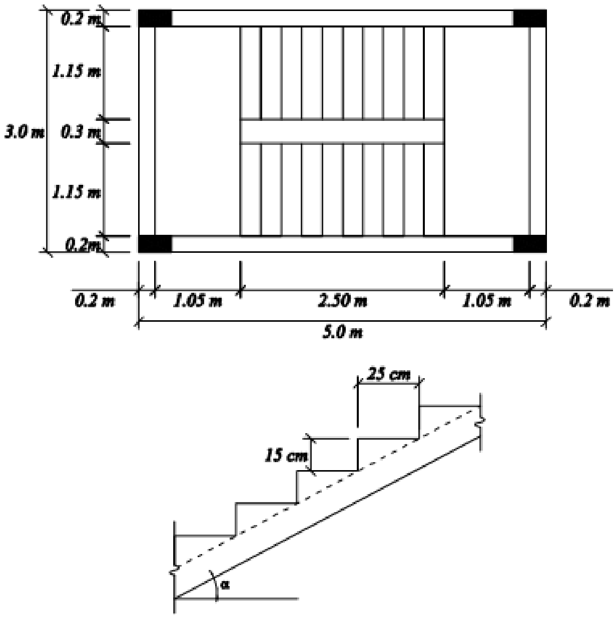

Figure 1

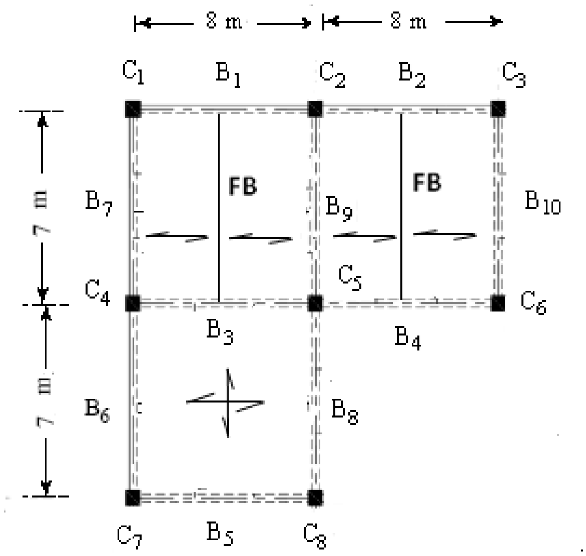

Figure 2

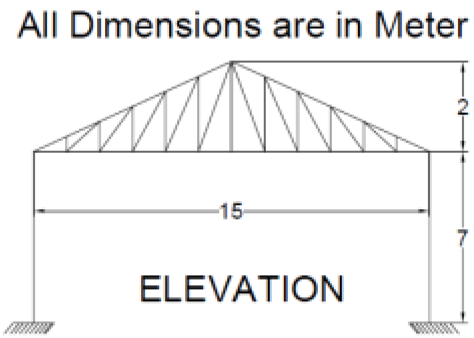

Figure 3

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